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An Autonomous Institute Affiliated to University of Mumbai

(Approved by AICTE, Accredited by NAAC 'A' Grade & Four Programs by NBA)

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“UNDERGROUND CABLE FAULT DETECTION SYSTEM”

Mini Project 2A

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Lokmanya Tilak College of Engineering

Third year Electrical Engineering

Semester V

Academic Year 2024-25

CERTIFICATE

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Declaration

We declare that the Mini Project report titled “UNDERGROUND CABLE FAULT DETECTION SYSTEM” is originally written by us and presenting our own idea. We have adhered to the Academic honesty and integrity in the preparation of the report. This Mini project report is not submitted anywhere else for any type of degree or award.

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ABSTRACT

Underground power cables are widely used in modern electrical distribution systems due to their safety, reliability, and aesthetic advantages over overhead networks. However, accurately locating faults in underground cables remains a major challenge because the cables are concealed beneath the ground and cannot be inspected visually. When a fault such as a short circuit, insulation failure, or line-to-ground defect occurs, it results in power disruption, voltage instability, and increased maintenance time. Traditional fault detection methods involve manual digging, time-consuming inspections, or the use of advanced but expensive equipment. To overcome these issues, the proposed project introduces a practical and cost-effective Underground Cable Fault Detection System that identifies the distance of the fault from the base station.

The system works on the principle of Ohm's Law, where a low DC voltage is applied through the cable, and variations in resistance are measured to determine the fault location. A microcontroller, preferably from the PIC family, is used to process the voltage drop across a series of resistors representing cable length. The data is then analyzed using the built-in ADC (Analog-to-Digital Converter), and the calculated fault distance is displayed on a 16x2 LCD screen. Fault creation is simulated using switches placed at fixed intervals, allowing accurate calibration and testing of the system. The hardware design incorporates essential components such as resistors, a voltage regulator, diodes, an LCD module, and a power supply unit.

This project provides a reliable, economical, and efficient approach to detecting underground cable faults with minimal human intervention. It reduces fault detection time, enhances maintenance accuracy, prevents excessive excavation, and supports uninterrupted power supply restoration. The concept can be further extended for real-time monitoring, automation, and integration into smart grid infrastructure. Overall, the system serves as a practical solution to one of the most critical issues in underground transmission network.

INTRODUCTION

Electric power distribution is essential for supporting domestic, commercial, and industrial needs. As cities expand and infrastructure becomes more complex, underground power cables have become a preferred option over traditional overhead systems. Unlike overhead lines, underground cables are better protected from environmental factors such as wind, rain, storms, falling branches, and pollution. They offer improved safety, reduced risk of electrical hazards, and a cleaner appearance in urban areas. However, the biggest challenge with underground cabling is detecting and locating faults when they occur. Since the cables are not visible or easily accessible, identifying the exact fault location becomes time-consuming and costly without proper technology.

The project “Underground Cable Fault Detection System” aims to provide an efficient solution for determining the location of faults in underground cables. The system is based on the principle of Ohm’s Law, where the change in resistance due to a fault is used to estimate the distance of the fault from the base station. By applying a low-voltage signal through a cable and analyzing the return current, the system can calculate the point at which a short circuit, line-to-ground fault, or insulation failure occurs. A microcontroller processes these voltage variations and displays the fault distance on an LCD screen in kilometers.

This project is beneficial for power distribution companies, maintenance teams, and engineers as it reduces downtime and improves fault detection accuracy. Instead of manually digging or using lengthy inspection methods, technicians can quickly identify the approximate location and begin repairs. The system’s design uses components such as a PIC microcontroller, resistors, an ADC module, switches, voltage regulators, and an LCD display. Overall, it provides a reliable, low-cost, and practical approach to solving one of the major challenges in underground power cable systems.

OBJECTIVE

The primary objective of this project is to design and develop a reliable, efficient, and cost-effective underground cable fault detection system capable of accurately determining the exact distance of a fault from the base station. The system operates on the principle of Ohm's Law, which relates voltage, current, and resistance to calculate the location of faults within a cable network.

In modern power distribution systems, underground cables are increasingly used in urban and industrial areas because they are safer, more aesthetic, and less affected by environmental conditions compared to overhead lines. However, locating faults in underground cables presents a significant challenge, as the faults are not visible and traditional methods often require manual digging and inspection, which are time-consuming, costly, and labor-intensive.

This project aims to overcome these challenges by implementing a microcontroller-based system integrated with Analog-to-Digital Converter (ADC) inputs, a resistor network to simulate the cable resistance per unit length, and an LCD display to show real-time fault distance. When a fault such as a short circuit, line-to-ground fault, or insulation breakdown occurs, the system measures the change in resistance and calculates the corresponding distance of the fault from the base station.

The proposed system ensures quick fault identification, thereby significantly reducing the power downtime and maintenance costs. It enhances the overall efficiency and reliability of power distribution networks, especially in densely populated or industrial regions where power stability is critical. Furthermore, the system's compact design, low cost, and user-friendly interface make it suitable for practical implementation and educational demonstration purposes.

By accurately detecting and locating faults, this project contributes to improving the safety, stability, and maintenance efficiency of underground power transmission systems, supporting the modernization of electrical infrastructure in both rural and urban areas.

LITERATURE SURVEY

Paper Title: Microcontroller-Based Underground Cable Fault Distance Locator

Author: Dr. R. K. Sharma, 2016

Explanation: This paper explains how a PIC microcontroller is used to measure fault distance using Ohm's Law. The author designed a setup with resistors representing cable length. The microcontroller calculates the fault location and displays the output on an LCD. This approach inspired the use of ADC and LCD modules in similar academic projects.

Paper Title: Resistive Fault Detection in Underground Power Cables Using Voltage Drop Analysis

Authors: S. Banerjee and P. Dutta, 2017

Explanation: Presented at an IEEE conference, this study focuses on resistance variation in faulty cable segments. It proved that the fault point can be estimated by detecting voltage drops along the cable. The concept strongly supports using resistor networks and Ohm's Law in fault detection systems.

Paper Title: Comparative Study of Underground Cable Fault Detection Techniques

Author: Prof. Meera Kulkarni, 2020

Explanation: This journal article evaluates different methods like time-domain reflectometry, GSM modules, and resistance-based measurement. It concludes that microcontroller-based systems are more cost-effective and suitable for academic and small-scale practical applications.

Underground Cable

An underground cable is an insulated electrical conductor installed below the surface of the ground to transmit and distribute power safely and efficiently. Unlike overhead lines, these cables are protected from environmental disturbances such as rain, wind, storms, and physical damage, making them ideal for urban areas, industrial zones, and high-density regions where safety, space utilization, and aesthetics are important. Underground cables consist of conducting materials like copper or aluminum, covered with layers of insulation, shielding, bedding, and protective sheathing to withstand moisture, soil pressure, chemical reactions, and thermal conditions. They are commonly used in power distribution networks, telecommunication systems, railways, and underground installations where reliability and long-term performance are essential.

Types of Underground Cable

1. Paper Insulated Lead Sheathed Cable (PILC)

These cables use layers of paper impregnated with insulating oil, covered by a lead sheath for protection. They were widely used in earlier power distribution systems due to their durability and mechanical strength, though they are less common today.

2. Cross-Linked Polyethylene (XLPE) Cable

XLPE cables are made with cross-linked polyethylene insulation, which offers high thermal resistance, excellent dielectric strength, and better current-carrying capacity. They are now the most commonly used cables in modern underground power networks.

3. Polyvinyl Chloride (PVC) Insulated Cable

PVC cables are cost-effective and suitable for low and medium voltage applications. They provide good insulation, resistance to moisture and chemicals, and flexibility in installation.

4. High Tension (HT) Cables

These cables are designed for medium to high voltage transmission ranging between 11kV and 33kV. They are used in industrial networks and large-scale underground power systems where higher load capacity is required.

5. Low Tension (LT) Cables

LT cables operate below 1kV and are mainly used in residential and commercial distribution. They are easier to install and maintain compared to high-voltage cables.

6. Gas-Insulated Cables (GIL)

These cables use inert gas, such as SF₆, as an insulator instead of solid materials. They are typically used in specialized applications where high reliability and minimal losses are required.

7. Oil-Filled Cables

These cables have oil-impregnated insulation with pressurized oil channels that prevent void formation and improve heat dissipation. They are mainly used in high-voltage transmission networks.

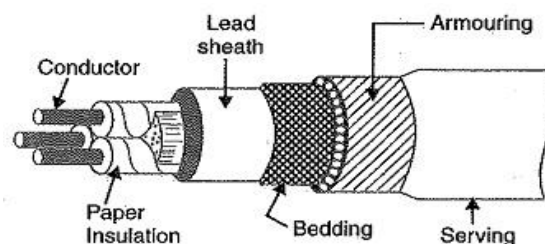
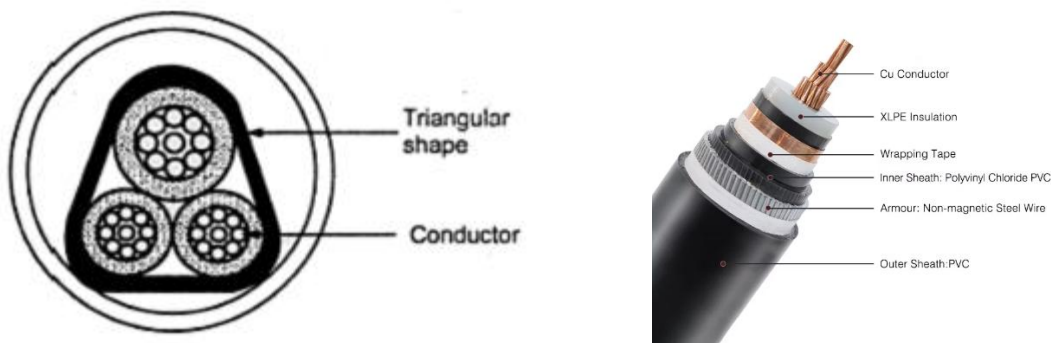


Fig. 11.1 Construction of a Cable

Underground Fault

An underground fault refers to any abnormal electrical condition that disrupts the normal flow of current in underground power cables, caused by issues such as insulation failure, conductor damage, moisture ingress, chemical corrosion, or mechanical stress from excavation or soil movement. These faults can result in short circuits, open circuits, or earth leakage, leading to power interruptions, voltage drops, or safety hazards. Unlike overhead cables, locating faults in underground systems is challenging because the cables are buried and not directly visible, requiring specialized detection methods and equipment. Effective fault identification is crucial to minimize downtime, reduce repair costs, and ensure continuous and safe power distribution.

Types of Underground Faults

1. Short Circuit Fault

This type of fault occurs when two conductors come into direct contact due to insulation failure or physical damage. It causes a sudden rise in current flow, which can damage equipment and disrupt power supply if not detected quickly.

2. Line-to-Ground Fault

In this fault, one of the phase conductors comes into contact with the earth or a grounded surface. It is one of the most common underground cable faults and can cause leakage current, voltage imbalance, and safety risks.

3. Open Circuit Fault

This happens when the conductor breaks or gets disconnected at some point along the cable due to mechanical stress, corrosion, or joint failure. As a result, the electrical circuit becomes incomplete, and power transmission stops beyond the fault point.

4. Double Line-to-Ground Fault

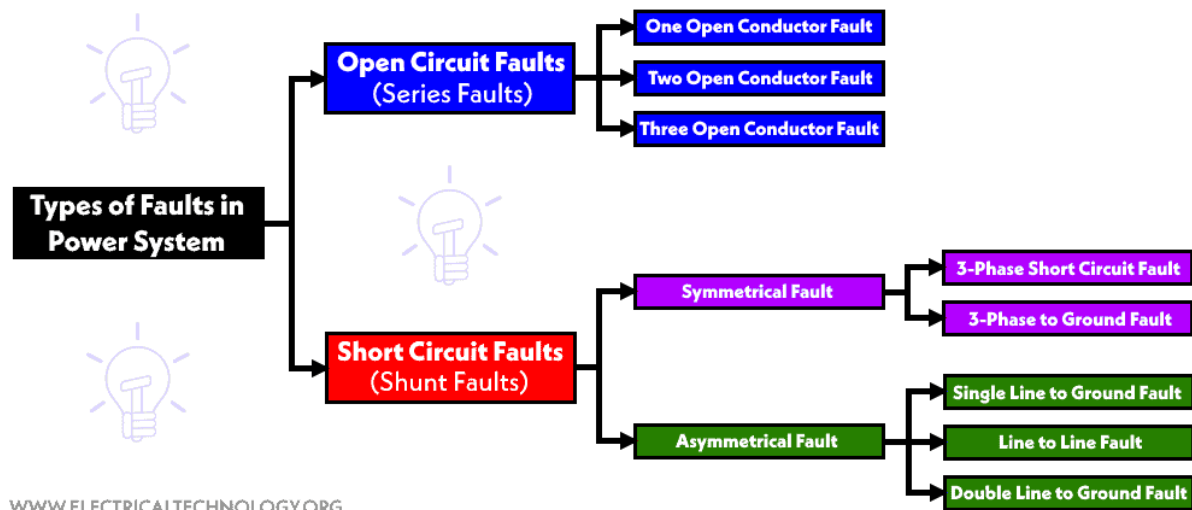
In this condition, two phase conductors come into contact with the ground simultaneously. It is more severe than a single line-to-ground fault and can cause significant power system instability.

5. Phase-to-Phase Fault

This type of fault occurs when two phase conductors touch each other due to insulation degradation or external damage. It leads to excessive current flow and overheating, which can damage cable components.

6. Symmetrical (Three-Phase) Fault

This is the least common but most severe fault, where all three phases are shorted together or connected to the ground. It causes a very high current surge and can lead to system-wide power failure if not controlled.



Working Principle

The underground cable fault detection system operates on the fundamental principle of Ohm's Law, which states that the resistance of a conductor is directly proportional to its length, provided the material and cross-sectional area remain constant. In underground power cables, when a fault such as a short circuit or line-to-ground fault occurs, the total resistance of the cable changes up to the point of the fault. By measuring this change in resistance, the system can accurately determine the distance of the fault from the base station. To achieve this, a low-level DC voltage is injected into the cable through a series of resistors, enabling the system to monitor the voltage variations that occur due to faults along the cable length. The DC voltage provides a stable and linear response, ensuring precise measurement of voltage drop along the line.

To demonstrate and test the system's functionality, slide switches are strategically placed at different intervals to simulate faults at specific distances. When a switch is turned on, it mimics a short circuit or ground fault at that point, allowing observation of system behavior under controlled conditions. As current flows through the resistive cable, a voltage drop occurs across each resistor in the network, and this drop depends on the fault's location. The nearer the fault to the base station, the lower the resistance and the higher the voltage drop across the affected section. These voltage variations are crucial in estimating the fault position accurately.

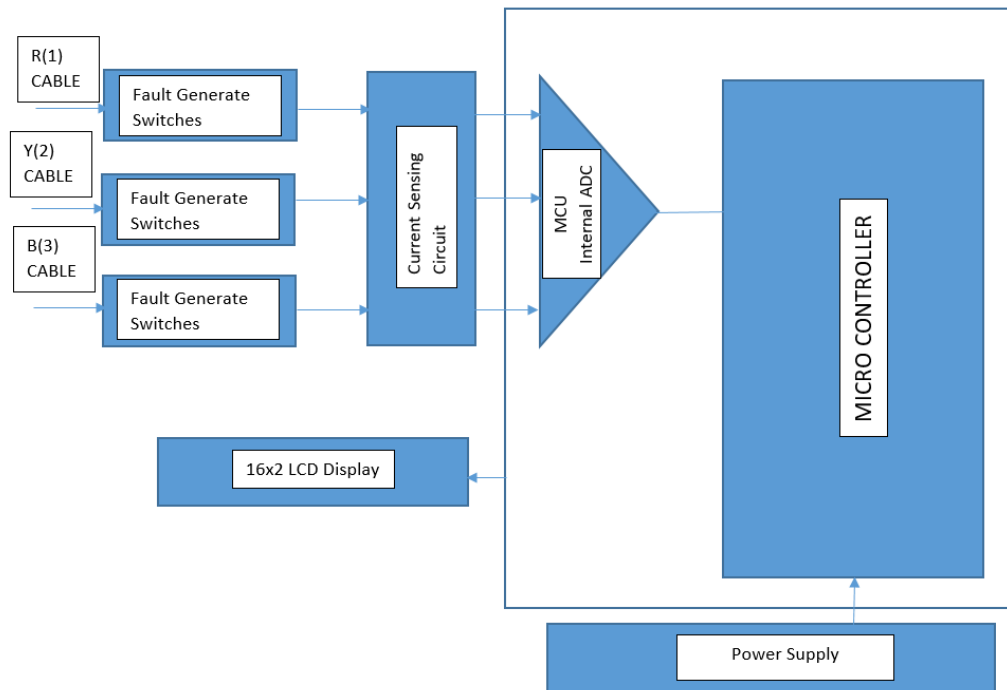
The varying analog voltages corresponding to different fault positions are then fed into the Analog-to-Digital Converter (ADC) of the PIC microcontroller, which converts them into digital data for further processing. The microcontroller, acting as the core processing unit, analyzes this digital data using pre-programmed logic to compute the fault distance based on the voltage-to-resistance relationship. Once the calculation is complete, the microcontroller converts the information into a human-readable form and displays the fault distance on a 16x2 LCD screen in meters or kilometers. This direct display helps maintenance engineers or technicians quickly identify and repair the fault without the need for manual calculations or time-consuming inspections.

A stable power supply is essential for the reliable operation of this system. Therefore, a

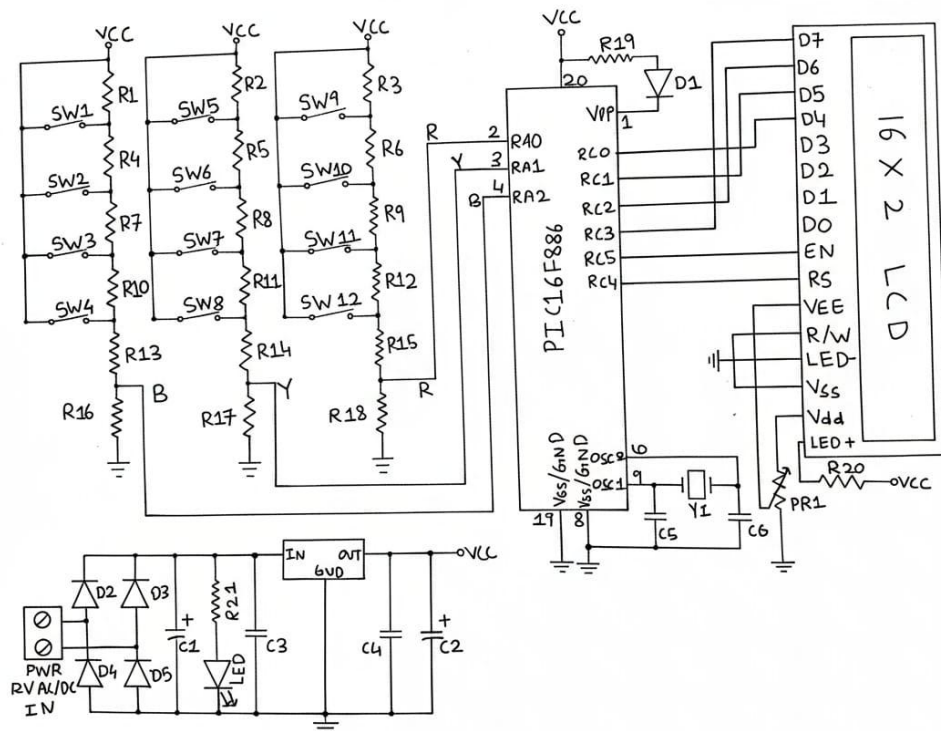
voltage regulator (LM7805) is used to maintain a constant 5V DC output, ensuring that the microcontroller and other components function smoothly even when input power fluctuates. Additionally, diodes are employed for reverse polarity protection, preventing accidental damage due to incorrect power connections, while capacitors are used for filtering electrical noise and stabilizing the power supply. These protection and stabilization features ensure the system's longevity and accurate performance under different electrical conditions.

In practical applications, this underground cable fault detection system offers significant advantages for power distribution networks, particularly in urban and industrial areas where underground cables are extensively used. It effectively reduces manual inspection efforts, shortens maintenance time, and ensures precise fault location, minimizing power outages and operational costs. By providing a quick, accurate, and automated way to detect and locate faults, the system enhances overall efficiency, safety, and reliability in electrical infrastructure. Furthermore, due to its simplicity, cost-effectiveness, and educational value, it serves as an excellent project model for students and researchers to understand real-world applications of microcontrollers and electrical fault detection principles.

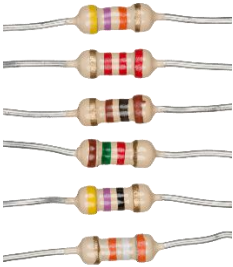
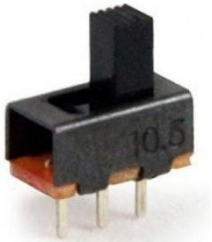
Block Diagram:

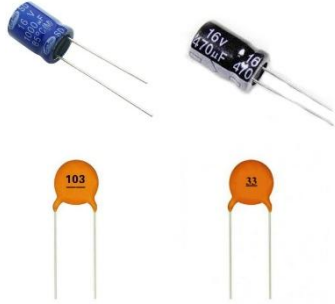


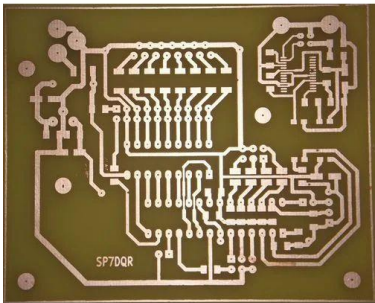
Circuit Diagram :-



LIST OF COMPONENTS

<u>SR NO</u>	<u>COMPONENTS</u>	<u>SYMBOL</u>
<u>1.</u>	PIC Microcontroller Acts as the main processing unit. It reads the analog input, processes the data using ADC, and calculates the fault distance. PIC16F886	
<u>2.</u>	Resistors Used to simulate cable lengths and create voltage drop variations according to fault positions. Different resistor values help in accurate measurement. 10K, 6K, 4K, 2K, 1K, 22E	
<u>3.</u>	Slide Switches These switches are used to simulate faults at different points along the cable path, allowing testing and calibration of the system.	
<u>4.</u>	16x2 LCD Display Displays the fault distance in kilometers and other important readings from the microcontroller for easy user interaction.	

<u>5.</u>	Voltage Regulator Provides a stable 5V DC output to power the microcontroller and connected components, ensuring smooth operation. LM7805	
<u>6.</u>	Rectifier Diodes Used for polarity protection and conversion of AC to DC supply where needed in the circuit. IN4007, IN4148	
<u>7.</u>	Capacitors (Electrolytic and Ceramic) Used for filtering, noise reduction, and voltage stabilization in the power circuit. 1000UF, 47UF, 0.1UF, 33PF	
<u>8.</u>	Crystal Oscillator Provides the required clock signal to the microcontroller for accurate timing and processing. 4MHz	
<u>9.</u>	LED Indicator Used to show the power status of the system, indicating whether the circuit is functioning correctly.	

<u>10.</u>	Power Supply Unit External power source that provides input to the regulator circuit, which then supplies power to all components. 9–12V AC/DC	
<u>11.</u>	Printed Circuit Board (PCB) Holds and connects all components in a compact and organized layout for stable and durable functioning.	

FUTURE SCOPE

- Integration of IoT technology for real-time monitoring and remote fault detection.
- Addition of GSM/GPS modules to send automatic fault location alerts to maintenance teams.
- Use of AI and machine learning for predictive fault analysis and preventive maintenance.
- Implementation of advanced sensors and microcontrollers to improve detection accuracy.
- Upgradation for smart grid and automation compatibility to ensure faster fault isolation and power restoration.

APPLICATIONS

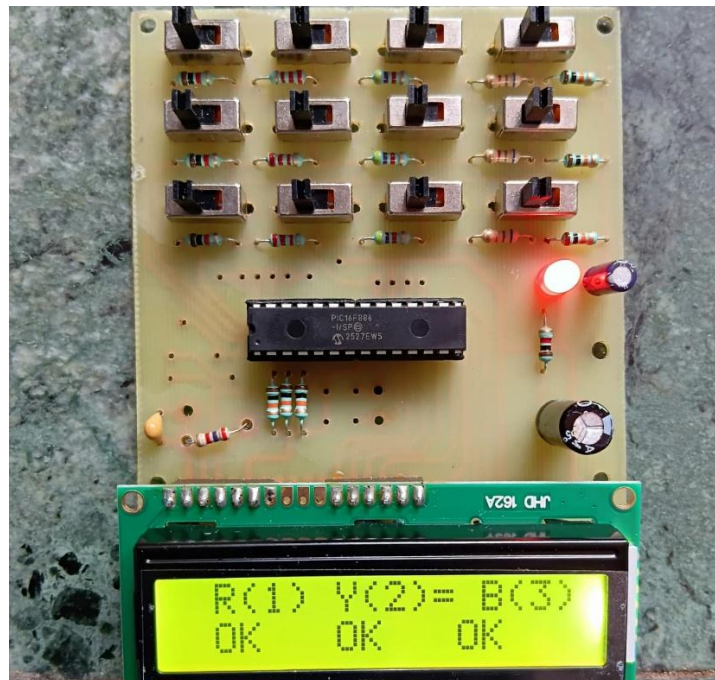
- Used in electrical power distribution systems for fault detection in underground cables.
- Suitable for industrial areas where underground cabling is used for safety and space saving.
- Helpful in smart city and smart grid projects for automated monitoring.
- Applied in metro stations and urban infrastructure where overhead wiring is not possible.
- Useful in educational and research institutions for demonstrating embedded system applications.
- Can be adapted for telecommunication or data cable networks to locate line faults.
- Supports maintenance teams in locating faults quickly and restoring power efficiently.

ADVANTAGES

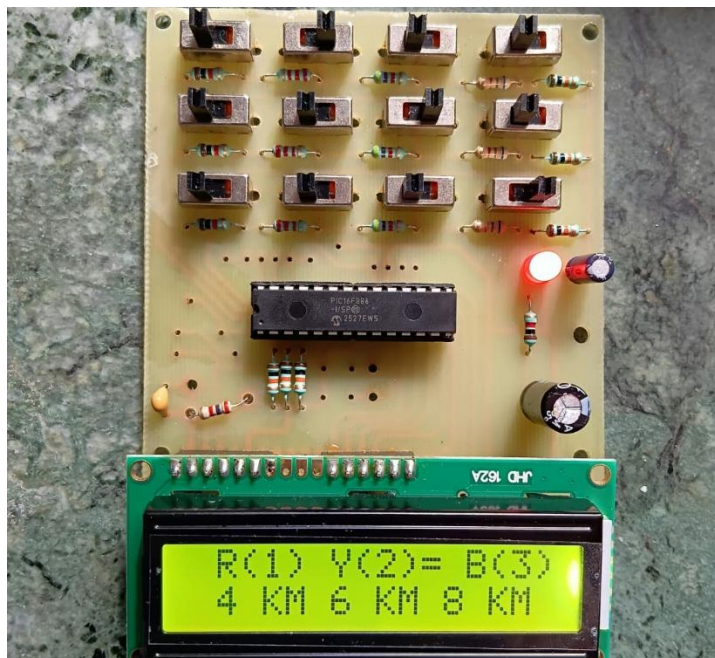
- Accurately detects the distance of underground cable faults.
- Saves time and effort during fault location and repair.
- Reduces overall maintenance and operational costs.
- Uses simple, low-cost, and easily available electronic components.
- Provides results instantly on an LCD display.
- Operates efficiently with low power consumption.
- Compact, reliable, and easy to install system.

RESULT

1. During Normal Conditions



2. During Faulty Conditions



Case Study :-

Case Description:

An MSEDCCL experienced a sudden power outage in a critical section of its production line. Upon investigation, it was discovered that a wire connecting an important component had burned, resulting in the disruption of the entire process. The burned wire was identified to be part of a larger circuit responsible for distributing high voltage to various machinery.

Repair Process - Compound Joint Application:

To address the issue, the maintenance team decided to repair the burned wire using a compound joint. A compound joint involves the use of specialized connectors or joints that are designed to withstand high temperatures and ensure a secure connection. The repair process consisted of the following steps:

1. The burned section of the wire was carefully cut and removed, leaving clean ends for reconnection.
2. A high-quality compound joint connector was chosen, specifically designed for high-temperature environments and secure wire connections
3. The compound joint was applied to connect the clean wire ends. The joint was crimped and sealed using a mold mixed with hardener to provide additional insulation and protection. Compound joint is mostly used for high voltage distribution.

Effectiveness of Repair:

Following the repair, the production line was gradually brought back online. The repaired wire and compound joint successfully restored the functionality of the circuit without any further interruptions. The compound joint demonstrated its capability to withstand high temperatures and maintain a secure connection, validating its effectiveness in this repair scenario.

Conclusion:

This case study highlights the importance of proper circuit design, maintenance, and repair techniques in preventing electrical faults and disruptions. The successful repair of a burned wire using a compound joint serves as an example of how innovative solutions can be employed to restore functionality and prevent potential hazards. It emphasizes the need for a proactive approach to electrical system maintenance and the adoption of suitable repair techniques to ensure the safety and reliability of industrial operations



CONCLUSION

The underground cable fault detection system provides an effective, low-cost, and practical solution to one of the major challenges in modern power distribution networks—locating faults in buried cables. By applying the principle of Ohm's Law and utilizing a microcontroller with ADC input, the system is capable of accurately detecting the distance of a fault and displaying it on an LCD. This reduces the time, cost, and effort required for manual inspections, while also minimizing downtime in power supply. The project demonstrates the integration of electronic components, embedded programming, and fault simulation to address real-world problems in electrical engineering. With further advancements, such systems can be enhanced using IoT, GSM, or wireless modules for remote monitoring, making them suitable for large-scale smart grid applications.

REFERENCES

- Paper Title: Microcontroller-Based Underground Cable Fault Distance Locator
Author: Dr. R. K. Sharma, 2016
- Paper Title: Resistive Fault Detection in Underground Power Cables Using Voltage Drop Analysis
Authors: S. Banerjee and P. Dutta, 2017
- Paper Title: Comparative Study of Underground Cable Fault Detection Techniques
Author: Prof. Meera Kulkarni, 2020